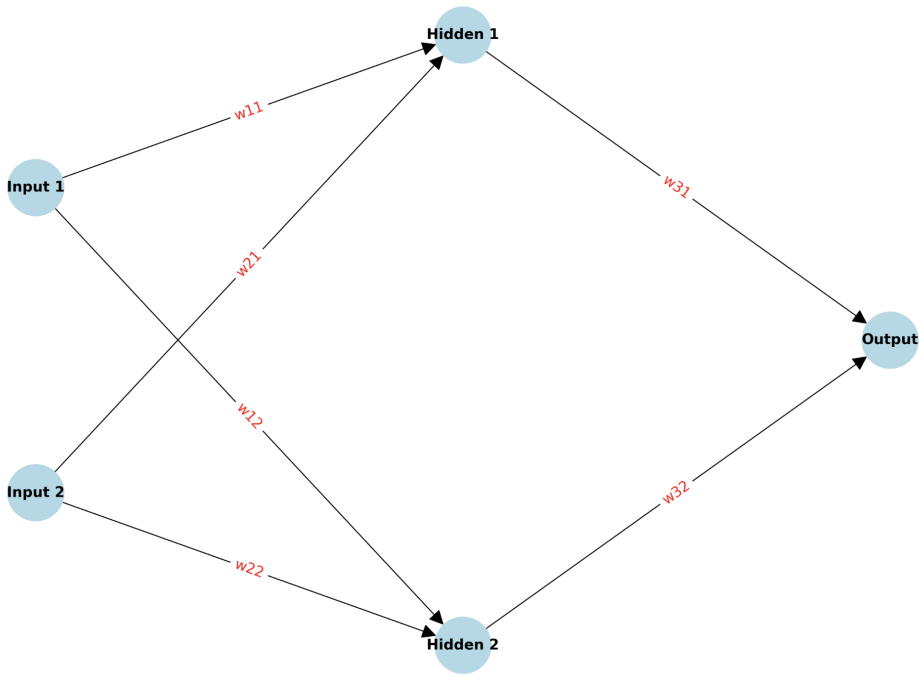
0.1 Multi-Layer Perceptron

Consider a two-layer perceptron with the following architecture and setup:



- **Input layer:** 2 neurons.
- **Hidden layer:** 2 neurons with Leaky-ReLU activation,

$$\text{LeakyReLU}(x) = \begin{cases} x & x > 0, \\ 0.01x & x \leq 0. \end{cases}$$

- **Output layer:** 1 neuron, also with Leaky-ReLU.
- **Initial weights:**

$$w_{11} = -0.20, \quad w_{12} = 0.40, \quad w_{21} = 0.10, \quad w_{22} = -0.50, \quad w_{31} = 0.60, \quad w_{32} = 0.30.$$

- **Loss:** for a prediction O and label Y,

$$J = \frac{1}{2}\big(O - Y\big)^2.$$

- **Learning rate:** $\alpha = 0.1$.

You are given *two* training examples

$$\big(X^{(1)}, Y^{(1)}\big) = \big([1 \quad -1],\ 0.5\big), \qquad \big(X^{(2)}, Y^{(2)}\big) = \big([2 \quad 3],\ 1.5\big).$$

1. **(5 points)** Compute the network output $O^{(1)}$ and $O^{(2)}$ for the two samples.
2. **(3 points)** Compute the individual losses $J^{(1)}$, $J^{(2)}$, and the average loss $\bar{J} = \frac{1}{2}\big(J^{(1)} + J^{(2)}\big)$.
3. **(8 points)** Use back-propagation to find $\nabla_w \bar{J}$ (the gradient of the *average* loss with respect to each weight $w_{11}, w_{12}, w_{21}, w_{22}, w_{31}, w_{32}$). Show your work.
4. **(4 points)** Perform *one* gradient-descent update with $\alpha = 0.1$ using the gradients from part (3). Report the new weight values.
5. **(5 points)** Suppose we instead chose $\tilde{\alpha} = 1$. Qualitatively compare $\tilde{\alpha}$ and $\alpha = 0.1$ for this network and data. Which would you prefer, and why?

Hints

- Chain rule for any weight W:
$$\frac{\partial J}{\partial W} = \frac{\partial J}{\partial O}\,\frac{\partial O}{\partial W}.$$
- Leaky-ReLU derivative:
$$\frac{d}{dx}\,\mathrm{LeakyReLU}(x) = \begin{cases} 1 & x > 0, \\ 0.01 & x \leq 0. \end{cases}$$